API Specification: Event-Driven Vector Vector Store Bridge

Endpoint base: `https://api.nexus-internal.net/v3/vector/sync` | **Auth:** `Bearer OAuth2 JWT (Scope: vector:write)` | **Rate Limit:** 5,000 req/min

1. OVERVIEW & INGESTION PROTOCOLS

The Vector Ingestion Bridge accepts real-time document streaming payloads, extracts structured metadata, processes semantic chunking via local GPU worker pools, and syncs vector indices to the primary storage cluster.

2. POST /V3/VECTOR/SYNC ENDPOINT

Submits raw unstructured payloads for contextual embedding generation and immediate indexing.

```
POST /v3/vector/sync HTTP/1.1 Host: api.nexus-internal.net Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9... Content-Type: application/json { "document_id": "DOC-2026-8841", "namespace": "hr_policies", "chunk_strategy": "parent_child", "chunk_params": { "parent_size": 1024, "child_size": 256, "overlap": 32 }, "metadata": { "classification": "internal", "owner_dept": "people_culture" } }
```

3. RESPONSE SCHEMAS & STATUS CODES

Successful execution returns HTTP 202 Accepted alongside a unique transaction reference for tracking asynchronous worker queue completion.

```
HTTP/1.1 202 Accepted Content-Type: application/json { "status": "QUEUED", "job_id": "job_99812_vec_sync", "chunks_created": 14, "estimated_completion_ms": 145 }
```

4. FAILURE RECOVERY & RETRY POLICIES

When vector indexing encounters lock contention, clients must implement exponential backoff retry cycles with a jitter constant ($k=1.5$). Maximum retry ceiling is set to 5 attempts before pushing payloads to the Dead Letter Queue (DLQ).